

HW3: Context Engineering — Rules & Process for Your P2 Project

Start Assignment

- Due Friday by 2:59am
- Points 40
- Submitting a file upload

Objective

Set up your P2 project for AI-assisted development by creating effective rules files and an Agile workflow. You'll bridge the gap between your PRD/mockups (Claude Web, Weeks 3-5) and IDE-centric development (Weeks 6+) by creating instructions that help your AI tools understand your project, follow Scrum practices, and start from your existing designs.

Prerequisites

Before starting this assignment, you should have:

- A PRD (from P1/P2 planning) with user stories and acceptance criteria
- At least one mockup or prototype (from Claude Web Artifacts or Figma)
- Your P2 repository created on GitHub
- Antigravity (or equivalent IDE AI tool) installed

Tasks

1. Set Up Scrum Workflow on GitHub (25%)

Create a professional Agile workflow using GitHub Issues and Projects:

- **Create a GitHub Project board** (Board layout) with columns: Backlog, Sprint Todo, In Progress, In Review, Done
- **Create milestones** for Sprint 1 and Sprint 2 (with due dates matching your P2 timeline)
- **Break your PRD into GitHub Issues** — Turn each user story into one or more issues with:
 - Clear title (action-oriented)
 - Acceptance criteria as a checklist in the issue body
 - Labels (`feature` , `bug` , `chore` , `docs` , priority labels)
 - Milestone assignment (Sprint 1 or Sprint 2)
- **Create at least 10 issues** across your two sprints
- **Assign Sprint 1 issues** to the "Sprint Todo" column

2. Create Rules / Instructions Files (35%)

Create a comprehensive `.antigravityrules` (or `CLAUDE.md` / `.cursorrules`) file for your P2 project that includes:

Project Context:

- Tech stack and versions
- Architecture overview (folder structure, key patterns)
- Naming conventions and coding standards
- Testing strategy (framework, patterns, coverage goals)

PRD & Design References:

- Link or reference to your PRD document
- Description of your mockup/prototype designs
- Key UI components and their expected behavior
- User flow descriptions that the AI should follow when implementing features

Scrum & Workflow Instructions:

- Branch naming convention (e.g., `feature/42-description`)
- Commit message format
- PR workflow (link to issues, required reviews)
- How to reference GitHub Issues in code and commits

Do's and Don'ts:

- Patterns to follow and patterns to avoid
- Dependencies and libraries to prefer
- Security and accessibility requirements

3. Test Your Rules with a Real Feature (25%)

Demonstrate that your rules file makes a measurable difference:

1. **Pick one feature** from your Sprint 1 backlog (a GitHub Issue)
2. **Implement it WITHOUT rules active** — Start a fresh chat, disable or remove the rules file, and ask the AI to implement the feature based on a brief description only
3. **Implement it WITH rules active** — Same feature, clean start, but with your rules file in place and referencing your PRD/mockups
4. **Compare the results** — Document specific differences in:
 - Code quality and consistency with your project patterns
 - How well the AI understood the design/mockup intent
 - Adherence to your naming conventions and architecture
 - Quality of tests generated

4. Refine and Document (15%)

- Iterate on your rules file based on the test results
- Document what worked, what didn't, and what you changed
- Write a brief reflection (1-2 pages) covering:
 - How your rules file bridges PRD/mockups to implementation
 - How the Scrum setup on GitHub helps organize AI-assisted development
 - What you would add or change for Sprint 2

Deliverables

Submit to Canvas:

1. **GitHub repository link** — With Project board, milestones, issues, and rules file visible
2. **Rules file** (`.antigravityrules` or equivalent) — In the project root, well-commented
3. **Before/after comparison** — Screenshots or code snippets showing the same feature implemented with and without rules
4. **Reflection document** (1-2 pages) — What you learned about context engineering for your own project

Grading Breakdown (40 points)

Category	Weight	Points
Scrum Setup	25%	10
Rules File Quality	35%	14
Before/After Test	25%	10
Reflection & Iteration	15%	6

Scrum Setup (10 pts)

- Project board with correct columns: 2 pts
- Milestones with due dates: 2 pts
- 10+ issues with acceptance criteria, labels, milestones: 4 pts
- Sprint 1 issues assigned to board: 2 pts

Rules File (14 pts)

- Project context (stack, architecture, conventions): 4 pts
- PRD/mockup references and design guidance: 4 pts
- Scrum/workflow instructions (branching, commits, PRs): 3 pts
- Do's and don'ts, well-organized: 3 pts

Before/After Test (10 pts)

- Clear methodology (same feature, clean conditions): 3 pts

- Specific, documented differences: 4 pts
- Measurable improvement demonstrated: 3 pts

Reflection (6 pts)

- Thoughtful analysis of what worked: 3 pts
- Evidence of iteration/refinement: 3 pts

HW3: Context Engineering — Rules & Process Rubric

Criteria	Ratings					Pts
Scrum Setup	10 pts Excellent — Project board, milestones, 10+ well- written issues from PRD	7.5 pts Good	5 pts Adequate	2.5 pts Needs Improvement	0 pts No Marks	10 pts
Rules File Quality	14 pts Excellent — Comprehensive, includes PRD/mockup references, workflow instructions	10.5 pts Good	7 pts Adequate	3.5 pts Needs Improvement	0 pts No Marks	14 pts
Before/After Test	10 pts Excellent — Clear methodology, measurable improvement with rules active	7.5 pts Good	5 pts Adequate	2.5 pts Needs Improvement	0 pts No Marks	10 pts
Reflection & Iteration	6 pts Excellent — Thoughtful analysis, evidence of refinement	4.5 pts Good	3 pts Adequate	1.5 pts Needs Improvement	0 pts No Marks	6 pts
Total Points: 40						